⇒ Inverse of Laplace

Ex) → $L\left\{ \frac{3}{s^2+4s+13} \right\}$

step: Make the Denominator in the form of perfect square.

$s^2+4s+13=0$

$s^2+4s=-13$

$s^2+4s+(2)^2=-13+(2)^2$

$(s-2)^2=-13+4$

$(s-2)^2=-9$

$\boxed{(s-2)^2-9=0}$

$L\left\{ \frac{3}{(s-2)^2-3^2} \right\}$

$\boxed{e^{2t}\cdot \sin 3t}$ Ans:- Prove it??

$\frac{1}{(s-2)}$ $e^{2t}\cdot \sin 3t$

$\frac{3}{s^2+3^2}$

$\boxed{\frac{3}{(s-2)^2+9}}$ Ans. proved.

⇒ Inverse Laplace Quiz type Questions

Q3

(a) $f(s) = \frac{1}{s^2-5s+6}$ → find the inverse laplace

$\frac{1}{s^2-3s-2s+6}$

$\frac{1}{s(s-3)-2(s-3)}$

$\frac{1}{(s-3)(s-2)}$

$\frac{1}{(s-3)} + \frac{1}{(s-2)}$

Now.

$\frac{A}{(s-3)} + \frac{B}{(s-2)} = \frac{1}{(s-3)} + \frac{1}{(s-2)}$

$A(s-2)+B(s-3) = 1$

Put $s=2$.

$0 + B(2-3) = 1$

$-B = 1$

$\boxed{B = -1}$

$\boxed{\text{Put } s=3}$

$A(3-2) + 0$

$\boxed{A = 1}$

Now Replace.

$\frac{1}{s-3} - \frac{1}{s-2}$

$\boxed{e^{3t} - e^{2t}}$

Ans

$$(a-b)^2 \qquad s^2+2^2 \qquad s^2=-4$$
$$a^2-b^2 \qquad (s+2)^2 \qquad s=\sqrt{-4}$$
$$s^2+2s+4 \qquad s=\pm 2i$$

(b) $F(s) = \dfrac{s+4}{s(s-1)(s^2+4)}$

$\dfrac{A}{s} + \dfrac{B}{(s-1)} + \dfrac{Cx+D}{s^2+4}$ rrowl.

$$\frac{s+4}{s(s-1)(s^2+4)} = \frac{A}{s} + \frac{B}{(s-1)} + \frac{Cx+D}{(s^2+4)}$$

$s+4 = \boxed{A(s-1)(s^2+4) + Bs(s^2+4) + (Cx+D)(s-1)(s)}$

~~$s+4 = A_0 - A$~~ Now put $s=1$

$s+4 = \underbrace{A(1-1)(1^2+4)}_{0} + B(1)(1^2+4) + \underbrace{(Cx+D)(1-1)(s)}_{0}$

$s+4 = B(5)$

$1+4 = 5B$

$\frac{5}{5} = B$

$\frac{1}{1} = B$

$\boxed{B=1}$

$A(0-1)((0)^2+4) + B(0)(0^2+4) + Cx+D(0-1)(0)$

$A(-4) + (0) + 0 = s+4$

$-4A = 0+4$

$\boxed{A=-1}$

This goes bail more forword another method.

→ Practice Question for Laplace inverse.

→ find $L^{-1}\left[\frac{1}{s(s+1)(s+2)}\right]$

sol

$$\frac{1}{s} + \frac{1}{(s+1)} + \frac{1}{(s+2)}$$

$$\frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+2)} = \frac{1}{s(s+1)(s+1)}$$

$$\boxed{A(s+1)(s+2) + Bs(s+2) + Cs(s+1) = 1}$$

Now put $s = -1$

$$A(-1+1)(-1+2) - B(-1+2) + C(-1)(-1+1) = 1$$

$$-B - 0 = 1$$

$$-B = +1$$

$$\boxed{B = -1}$$

$\boxed{\text{Now } s = -2}$

$$A(-2+1)(-2+2) + B(-2)(-2+2) + C(-2)(-2+1) = 1$$

$$0 + 0 - 2C = 1$$

$$\boxed{C = +\frac{1}{2}}$$

$\boxed{s = 0}$

$$A(0+1)(0+2) + B(0)(0+2) + C(0)(0+1) = 1$$

$$2A = 1$$

$$\boxed{A = \frac{1}{2}}$$

Now. Replace

→

$$\frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+2)}$$

$$\frac{\frac{1}{2}}{s} + \frac{-1}{(s+1)} + \frac{\frac{1}{2}}{(s+2)}$$

$$\frac{1}{2} L^{-1}\left[\frac{1}{s}\right] - 1\, L^{-1} \frac{1}{(s+1)} + \frac{1}{2} L^{-1}\left[\frac{1}{s+2}\right]$$

$$\frac{1}{2}(1)(-1)e^{-t} + \frac{1}{2}e^{-2t}$$

$$\boxed{\frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}}$$ Ans,